JOSÉ MANUEL REQUENA PLENS

Embedded Firmware & Software Engineer
C · C++ · Python · RTOS · Test Automation · CI/CD

jmrplens@gmail.com | Valencia, Spain | jmrp.io | linkedin.com/in/jmrplens
github.com/jmrplens | Google Scholar

For more information you can click on the name of places, certificates or courses that appear in the curriculum.

WORK EXPERIENCE

Software Engineer

Power Electronics

April 2023 – May 2026 | R&D Department. Solar area.

Firmware development for solar inverters and control systems in photovoltaic plants, working on STM32 microcontrollers with bare-metal and RTOS environments.

- Optimized the embedded file system, reducing access times and improving stability under continuous operation.
- Improved TCP communications performance, reducing latency and increasing the robustness of the network stack.
- Optimized tasks on STM32: reduced stack and heap usage, and tuned CPU cycles to improve real-time performance.
- Implementation and maintenance of industrial communications: Modbus RTU/TCP, UART, I2C and SPI.
- Development in C with HAL layers and low-level peripheral management.
- Hardware-in-the-loop (HIL) test automation in Python (pytest): unit, integration and stress tests integrated into CI/CD pipelines.

Predoctoral researcher

Polytechnic University of Valencia + CSIC + Hospital La Fe + AVI

January 2023 – April 2023 | Medical Imaging and Therapy Systems i3M

Researcher in the R&D line: **Acoustic Metamaterials for Ultrasound Histotripsy treatments**.
Project: New generation of smart metasurfaces based on additive manufacturing for strategic applications in telecommunications (Metasmart). Funded by the Valencian Agency for Innovation (Reference: INNEST/2022/345).

- Ultra-close focus acoustic lens design.
- Prototype manufacturing.
- Experimentation with ex-vivo tissues.

Predoctoral researcher

Polytechnic University of Valencia + CSIC

July 2021 – July 2022 | Medical Imaging and Therapy Systems i3M

The main objective is to conduct research and development of acoustic metamaterials for use in architectural applications and medical imaging.
Goals achieved:

- Theoretical development of acoustic metadiffusers based on membranes or plates reducing the size of commercial diffusers.
- Simulation of the acoustic diffusers and validation of results (MATLAB & COMSOL).
- Two publications in national and international congresses:
  - Tecniacústica - **Beyond Schroeder diffusers using acoustic metasurfaces**
  - Euronoise - **Sound diffusing metasurfaces based on elastic plates and membranes**
- Development and simulation of metamaterial-based waveguides for improved ultrasound resolution in air.
- Fabrication of waveguide prototypes.
- Validated experimental results against those obtained in simulation.
- Advisor of the Master thesis of a student of the Master's Degree in Acoustic Engineering of the UPV.

Researcher in European project

Higher Polytechnic School of Gandia

February 2020 – July 2021 Complex Media Acoustics Group of IGIC

Implement and experimentally validate a sound mitigation method, applicable to a real space vehicle launch configuration (VEGA), that results in a significant decrease in sound pressure levels generated in the launch area during spacecraft liftoff.

Project funded by ESA (European Space Agency) with reference ESA AO/1-9479/18/NL/LvH, in collaboration with: CNRS/Laboratoire d'Acoustique de la Université du Mans; COMET Ingeniería; Spanish National Research Council (CSIC) and Polytechnic University of Madrid.

All project objectives were completed at 100%:

- Design of the optimum geometry of the acoustic metamaterial to reduce noise levels.
- Simulation of a real environment to validate design performance (MATLAB, COMSOL & Python).
- Design of the industrial model for the manufacture of the prototype by means of plastic injection.
- Development of software for acoustic measurements according to ISO 10534-2:1998 and ASTM 2611-19 standards. **A|Lab**.
- Design and development of a 3-axis robotic system to perform experimental measurements. Photos and some information here: **NEWS**.
- Results validated and accepted by ESA.
- Three publications/presentations at international congresses (Euronoise and ECSSMET) and one at a national congress (Tecniacústica):
  - Euronoise - **Perfect broadband sound absorber metamaterial for noise reduction in a rocket launch**
  - Euronoise - **Application of metamaterials to control noise scattering during space vehicle lift-off**
  - ECSSMET - Launch sound level characterisation and mitigation: numerical modelling framework and metamaterial proof of concept
  - Tecniacústica - **Acoustic field prediction during the launch of rockets**

Research internship

Dep. of Physics, Systems Engineering and Signal Theory

February 2018 – July 2018 University of Alicante

Internship in different research projects in the **Applied Acoustics Group**, belonging to the University Institute of Physics Applied to Sciences and Technologies of the Higher Polytechnic School of Alicante.
Goals achieved:

- Simulation of acoustic models.
- Acoustic measurements by means of a LabView-controlled robotic system.
- Investigation of the acoustic radiation efficiency (Vibration-borne noise) of metallic plates for an initial study of ship noise emissions in water. Carried out in collaboration with SAES, here you can read a news item on the subject: **NEWS**.
- A publication at a national congress:
  - Tecniacústica - **Comportamiento vibroacústico de contenedores cilíndricos en aire**

Founder and Vocal Member

AμTech

September 2016 – July 2018 University of Alicante

Association based at the Higher Polytechnic School of the University of Alicante. Founded in 2016. Promotes project-oriented microcontroller programming knowledge.
Activities performed:

- Group classes of reinforcement and extension of the subject "Digital Electronic Systems" of the Degree in Sound and Image Engineering of the University of Alicante.
- Laboratory organization and planning (located at the Colegio Mayor).
- Communication and Public Relations.

Sound technician

Acusticox

June 2011 – February 2018 | Cox, Alicante

Installation and tuning of audio equipment for live events and permanent installations; FOH technician.

OTHER KNOWLEDGE AND SKILLS

Embedded Firmware & Electronics

C	●●●●○
STM32 & ARM	●●●●○
ESP32 / ESP-IDF	●●●●○
PlatformIO	●●●●○
Texas Instruments	●●●○○
FreeRTOS / RTOS	●●●●○
Comm. Protocols	●●●●○
HW Debugging	●●●●○
Keil uVision	●●●●○
Secure Boot & Embedded Sec.	●●●○○
Doxygen	●●●●○
Instrumentation	●●●●○

Quality, CI/CD & DevSecOps

GitHub Actions	●●●●○
GitLab CI	●●●○○
Jenkins	●●○○○
SonarQube / SonarCloud	●●●●○
GoReleaser	●●●○○
Static Analysis & SAST	●●●○○
Testing	●●●●○
pre-commit hooks	●●●●○

Networking

TCP/IP & Networking	●●●●○
MikroTik RouterOS	●●●●○
WireGuard / VPN	●●●●○
IPv6 & Dual-stack	●●●●○
QUIC / HTTP3	●●●○○
DNS / Dynamic DNS	●●●○○

Software Engineering & Tools

Python	●●●●○
C++	●●●○○
Go	●●○○○
TypeScript	●●○○○
Bash	●●●○○
Clang Tooling	●●●○○
Git	●●●●○
GitHub	●●●●○
GitLab	●●●●○
Docker	●●●●○
MATLAB	●●●●○
CMake	●●●○○
JetBrains Toolbox	●●●●○

Security & Cryptography

Applied Cryptography	●●●○○
TLS / mTLS / PKI	●●●●○
Nginx Hardening	●●●●○
CrowdSec	●●●○○
Network Hardening	●●●●○

AI & Agentic Engineering

Model Context Protocol (MCP)	●●●●○
AI Agents & Best Practices	●●●●○
LLM Integration	●●●○○

Simulation & Acoustics

COMSOL Multiphysics	●●●●○
LabView	●●●○○
Acoustic Instrumentation	●●●●○

General & Platforms

Linux / Unix	●●●●○
MacOS	●●●●○
Windows	●●●●○
LaTeX	●●●●○
HTML / CSS	●●●○○

EDUCATION

Formal education

PhD in Health and Wellness Technologies

Polytechnic University of Valencia

September 2020 – 2023 (not completed) Medical Imaging and Therapy Systems i3M

Title Industrial and biomedical applications of metamaterials for the control of acoustic beams. In this project, we propose to combine new air-coupled ultrasound transducers with hyperbolic metamaterials to design vision systems that exceed the imaging capabilities of traditional systems.

Description Research in acoustic engineering applied to biomedicine: biomedical signal processing, ultrasound characterization, and computational modeling and simulation of acoustic phenomena. I redirected my career toward industrial software development before completing the doctorate, applying my foundation in programming, data analysis and analytical problem-solving.

Academic record

Master's Degree in Acoustical Engineering

Polytechnic University of Valencia

September 2018 – June 2019 Higher Polytechnic School of Gandia

Degree obtained with honors in: Fundamentals of Acoustics, Acoustic Isolation, Musical Acoustics, Signal Processing in Acoustic Engineering, Ultrasonics and Acoustic Simulation Techniques.

Final Master's thesis entitled "Difusores acústicos basados en resonadores de membrana y placa" ("Acoustic diffusers based on membrane and plate resonators"), qualified with honors with special mention.

Diploma Academic record

Degree in Telecommunication Engineering in Sound and Image

University of Alicante

July 2018 Higher Polytechnic School

Degree obtained together with the 2 specializations: Acoustic Engineering and Audiovisual Technology.

Final Degree Work entitled "Estudio de la relación campo directo/reverberado; útil/perjudicial" ("Study of the direct/reverberated field relationship; useful/harmful").

Diploma Academic record

Certificate of Higher Education in Electronics Technician & Superior Sound Technician

IES Salesianos (San José Artesano) | IES Luis García Berlanga + Ciudad de la Luz

2009 | 2011 Elche | Alicante

Diplomas

CERTIFICATES

The certificates can be accessed by clicking on the name of each one.

Occupational Health and Safety

Generic	INVASSAT. **50** H.
Nanomaterials	INVASSAT. **50** H.
Chemical sector	INVASSAT. **50** H.
Emergencies	INVASSAT. **70** H.
Nutrition	INVASSAT. **50** H.
Educational	INVASSAT. **50** H.
Services	INVASSAT. **50** H.
Researcher	UPV. **15** H.

Labor/Industrial

Op. industrial trucks	Gescoform. **15** H.
Food Hygiene Course	Asonaman. **30** H.
Self-protection plans	INVASSAT. **15** H.
Static electricity	INVASSAT. **15** H.
Data privacy	CSIRT-CV. **10** H.

Transversal competences

Gender perspective	EVES. **20** H.
Teamwork	Labora. **25** H.
Design Thinking	Labora. **25** H.
Critical thinking	Labora. **25** H.
Adaptability, flexibility	Labora. **25** H.
Autonomy, innovation	Labora. **25** H.
Prof. efficiency	Labora. **25** H.
Entrepreneurship	UPV. **20** H.
Gender persp. in research	UPV. **50** H.

Information technology

Using Python for Research	HarvardX. **50** H.
Analyzing Data w/ Python	IBM. **20** H.
Visualizing Data w/ Python	IBM. **20** H.
Numerical methods (MATLAB)	UPV. **50** H.
Scientific computing	UPV. **50** H.
Documents with LaTeX	UPV. **56** H.